

Loss Reserving Exhibits 1993–2024

A side-by-side breakdown of three claims reserving methods applied to the same paid-loss triangle — swipe through for development factors, paid triangles, and a full method comparison.

- 01 · Chain Ladder
- 02 · London Chain
- 03 · Taylor Separation



01

Chain Ladder

02

London Chain

03

Taylor Separation

04

Summary & Comparison

EXHIBIT 01

Chain Ladder Method

Projects future claim payments by applying age-to-age loss development factors, derived from the average historical progression between successive development periods, to the latest diagonal of paid losses.

DEVELOPMENT FORMULA

$$C_{i,k+1} = C_{i,k} \times f_k$$

WHERE

$$f_k = \Sigma C_{i,k+1} / \Sigma C_{i,k}$$

TOTAL ULTIMATE

MAD 3.24bn

MAD 3,244,834,777

PAID TO DATE

MAD 2.76bn

MAD 2,755,850,270

TOTAL IBNR

MAD 489.0m

MAD 488,984,507

IBNR / ULTIMATE

15.1%

reserve as % of total ultimate

Fig. 1.1 Age-to-age (bars) · Cumulative to ultimate (line)

Loss Development Factors

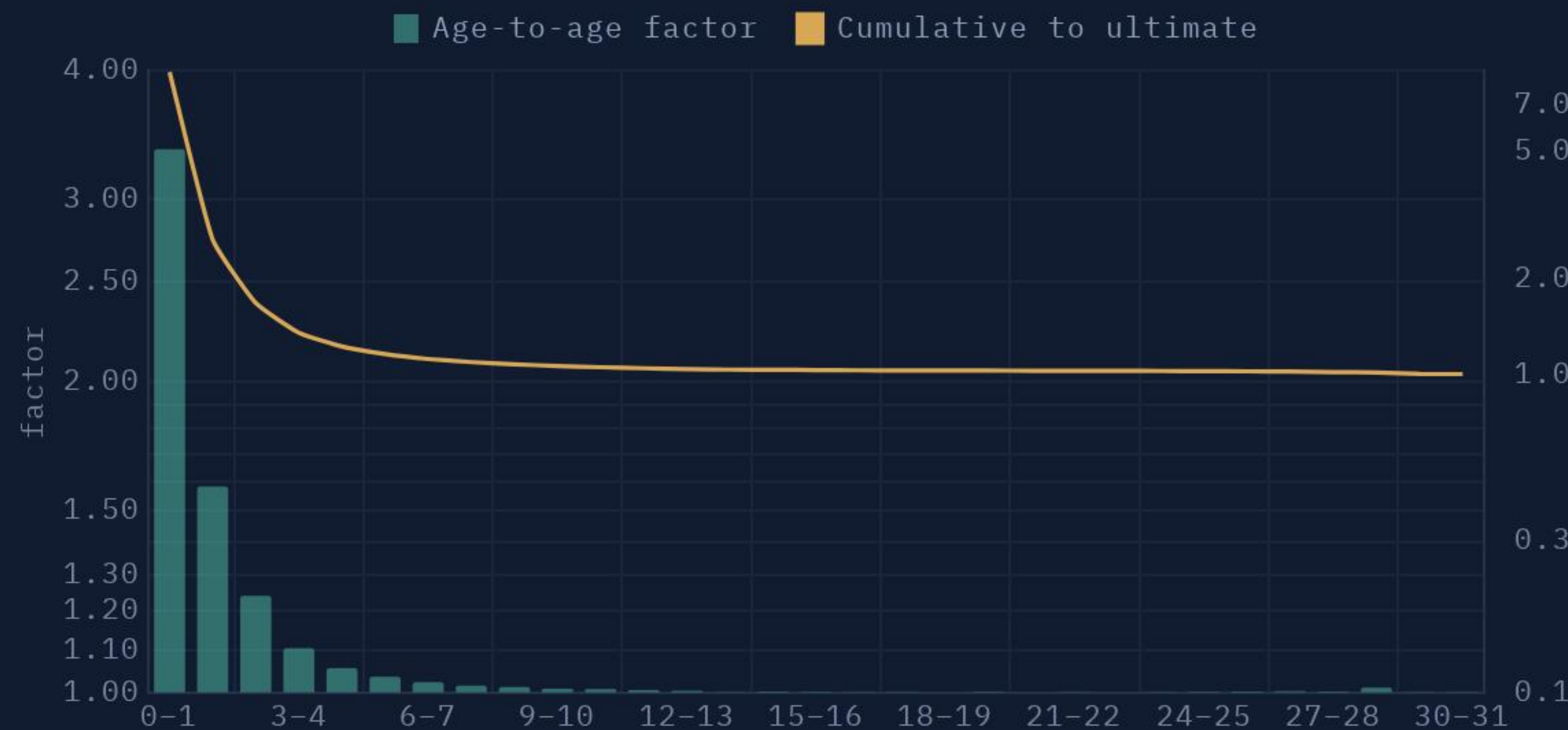


Fig. 1.2 % of ultimate by accident year

Cumulative Paid Triangle

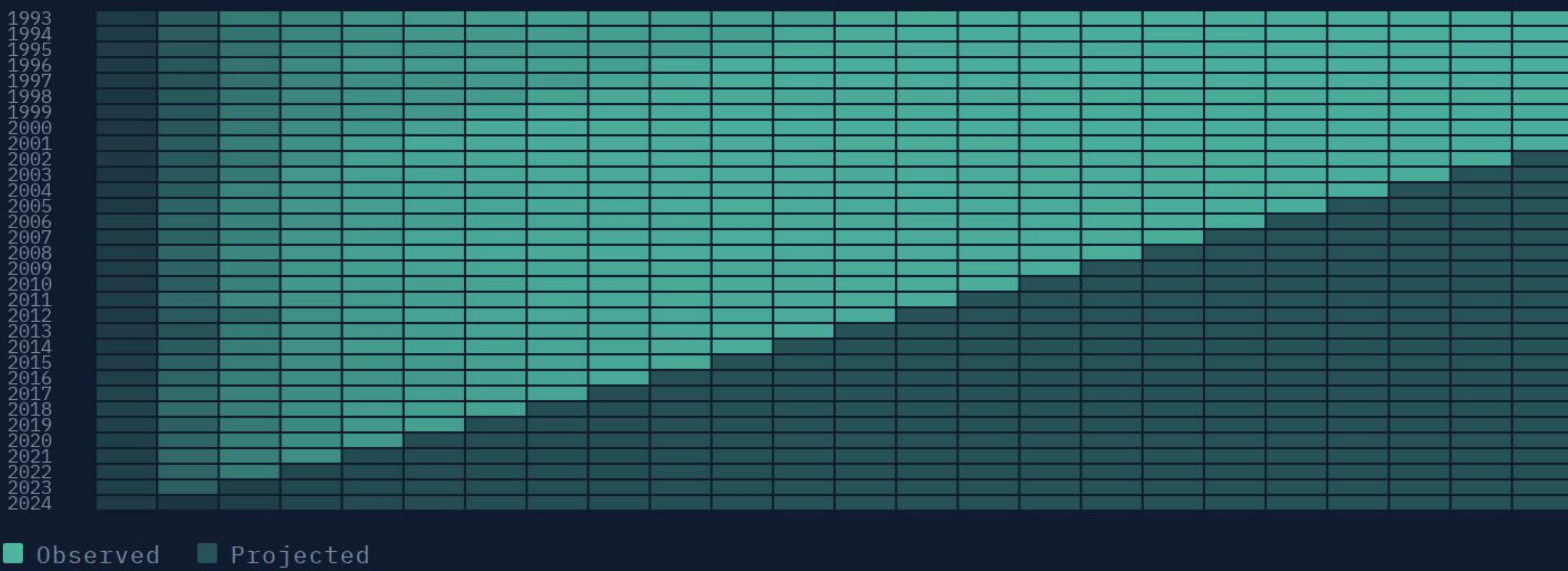


Fig. 1.3 Paid-to-date vs. reserve

Ultimate Loss by Accident Year



Fig. 1.4 MAD, undiscounted

IBNR by Accident Year





EXHIBIT 02

London Chain Method

A regression-based variant of the chain ladder: for each development period, cumulative losses at k+1 are fitted as a linear function of losses at k, allowing an intercept term to absorb a fixed cost component that a proportional (multiplicative) factor alone cannot capture.

DEVELOPMENT FORMULA

$$C_{i,k+1} = a_k + b_k \times C_{i,k}$$

WHERE

a_k , b_k = OLS intercept & slope

TOTAL ULTIMATE

MAD 3.17bn

MAD 3,165,387,299

PAID TO DATE

MAD 2.76bn

MAD 2,755,850,270

TOTAL IBNR

MAD 409.5m

MAD 409,537,029

IBNR / ULTIMATE

12.9%

reserve as % of total ultimate

Fig. 2.1

b_k coefficient

Regression Slope by Development Period



Fig. 2.2

% of ultimate by accident year

Cumulative Paid Triangle

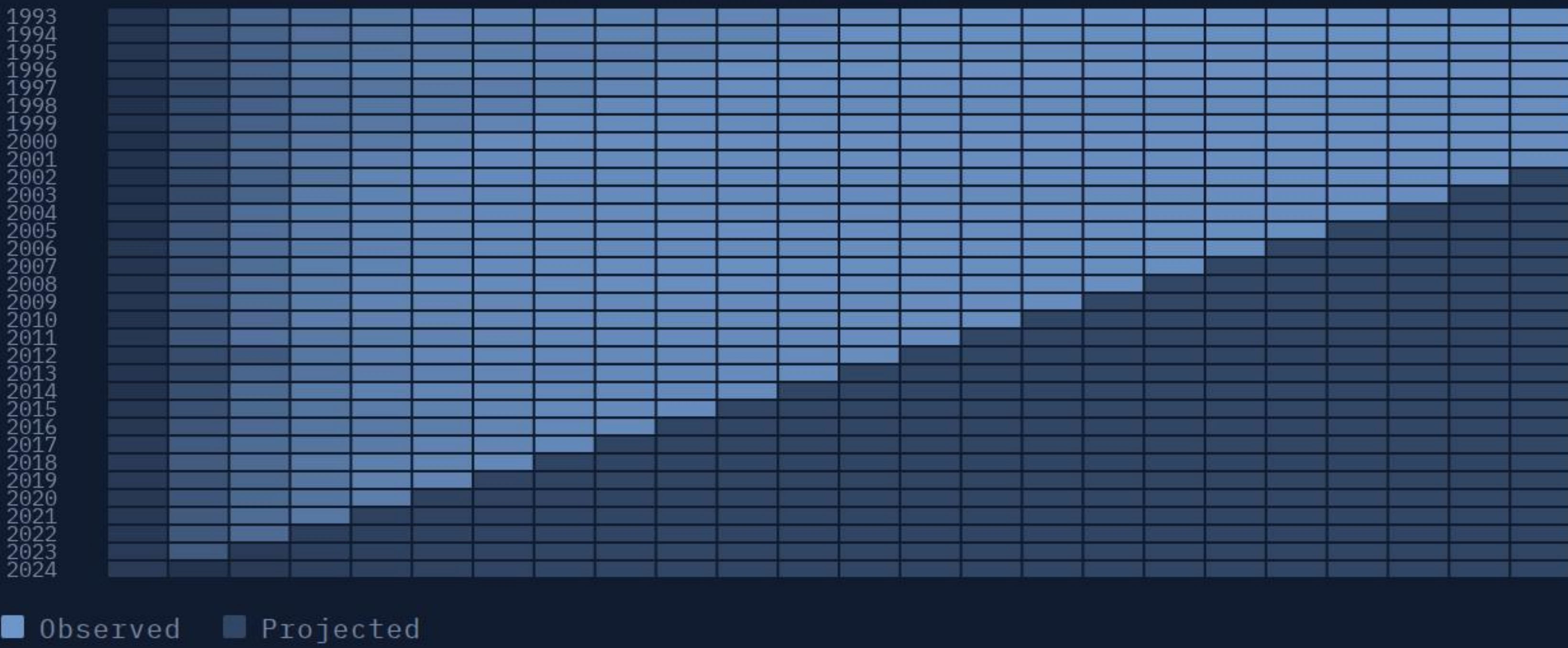


Fig. 2.3

Paid-to-date vs. reserve

Ultimate Loss by Accident Year



Fig. 2.4

MAD, undiscounted

IBNR by Accident Year





EXHIBIT 03

Taylor Separation Method

Decomposes incremental payments on the calendar-year diagonal into a settlement pattern $r(j)$, fixed by development age, and an inflation / claims-cost index $\lambda(t)$, fixed by calendar period — separating volume from monetary trend before projecting the future square.

SEPARATION FORMULA

$$X_{i,j} = r(j) \times \lambda(i+j)$$

FITTED TREND

$$\ln \lambda(t) = \alpha + \beta t$$

TOTAL ULTIMATE

MAD 3.13bn

MAD 3,125,577,120

PAID TO DATE

MAD 2.76bn

MAD 2,755,850,270

TOTAL IBNR

MAD 369.7m

MAD 369,726,850

IMPLIED ANNUAL TREND

1.5%

fitted $\lambda(t)$ inflation rate

Fig. 3.1 Calendar trend · log scale, with fitted line

Inflation Index $\lambda(t)$

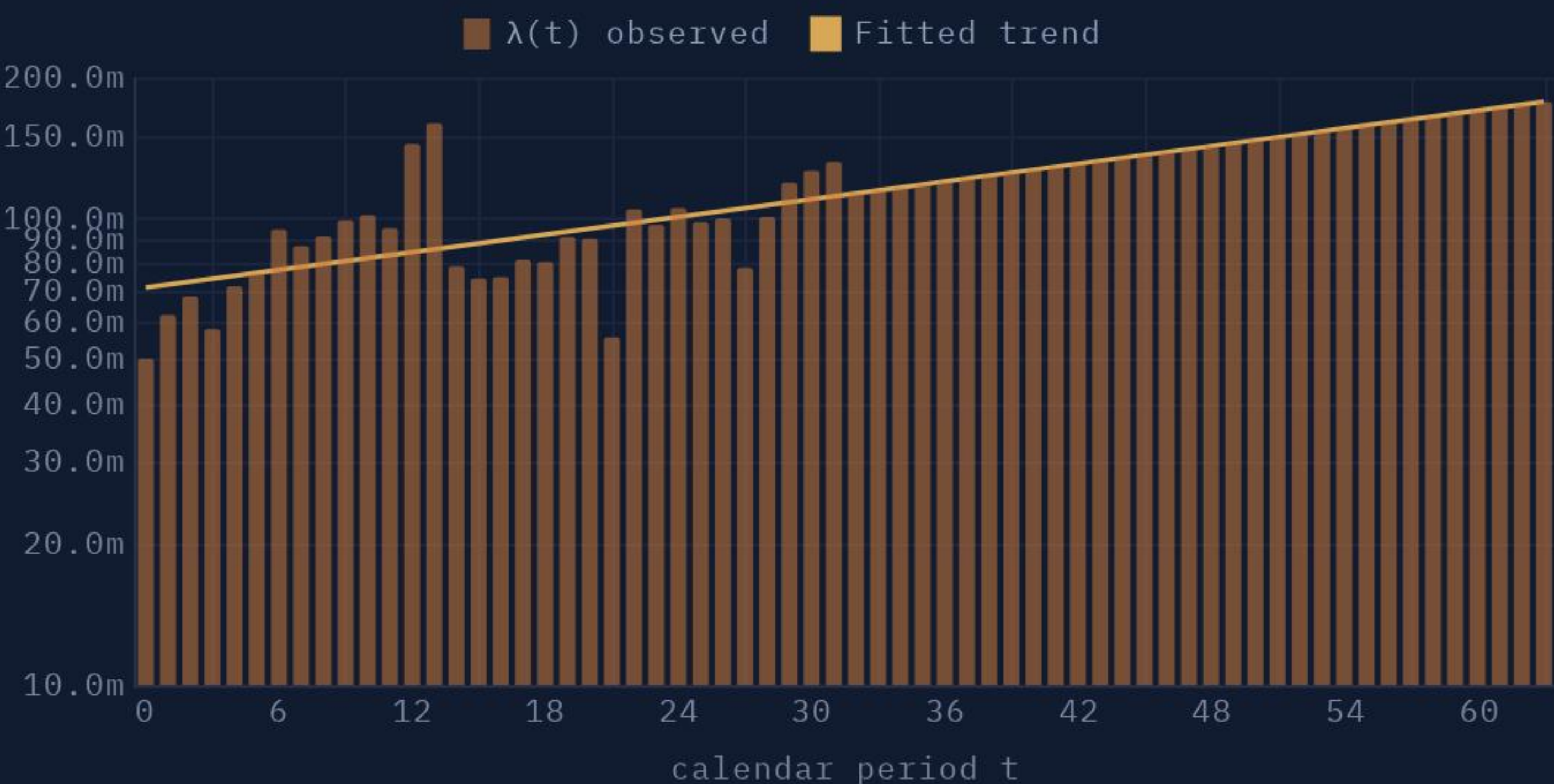


Fig. 3.2 % of ultimate by accident year

Cumulative Paid Triangle

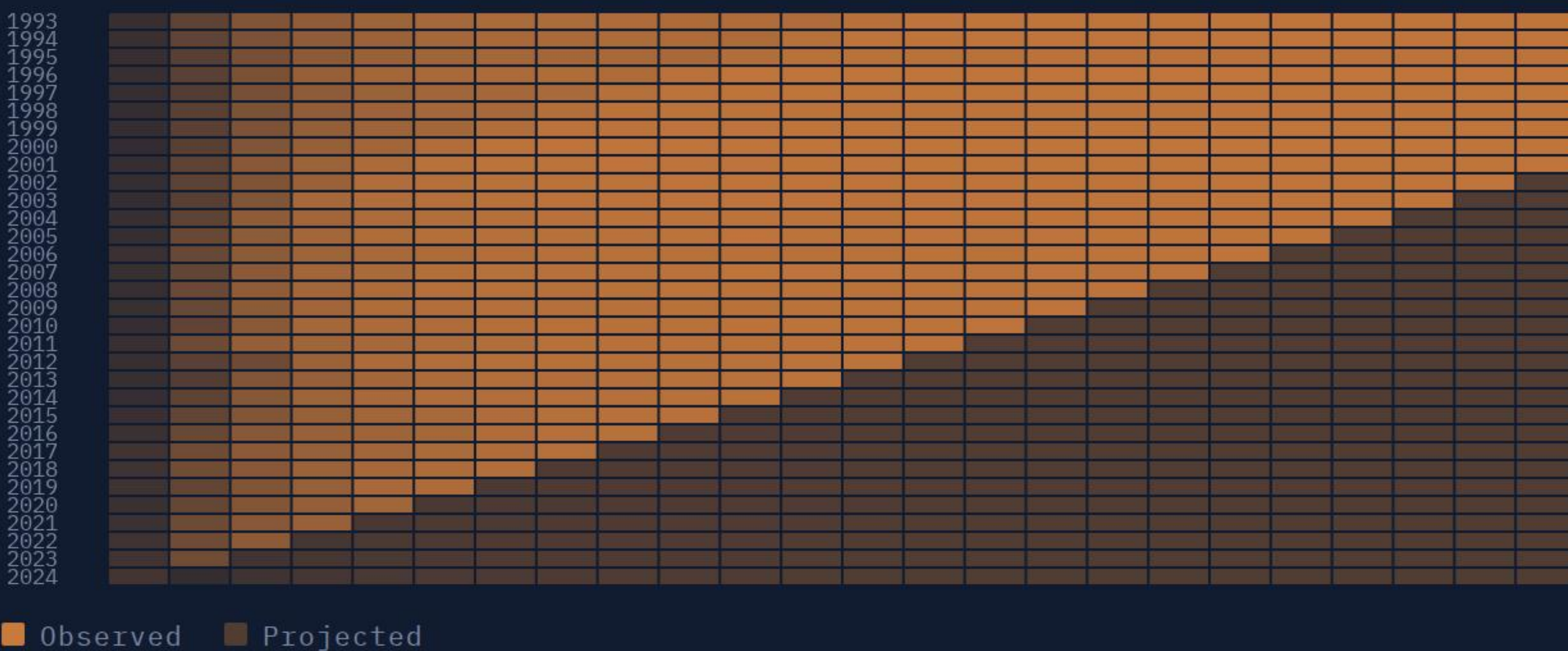


Fig. 3.3 Share of ultimate paid by development age

Settlement Pattern $r(j)$

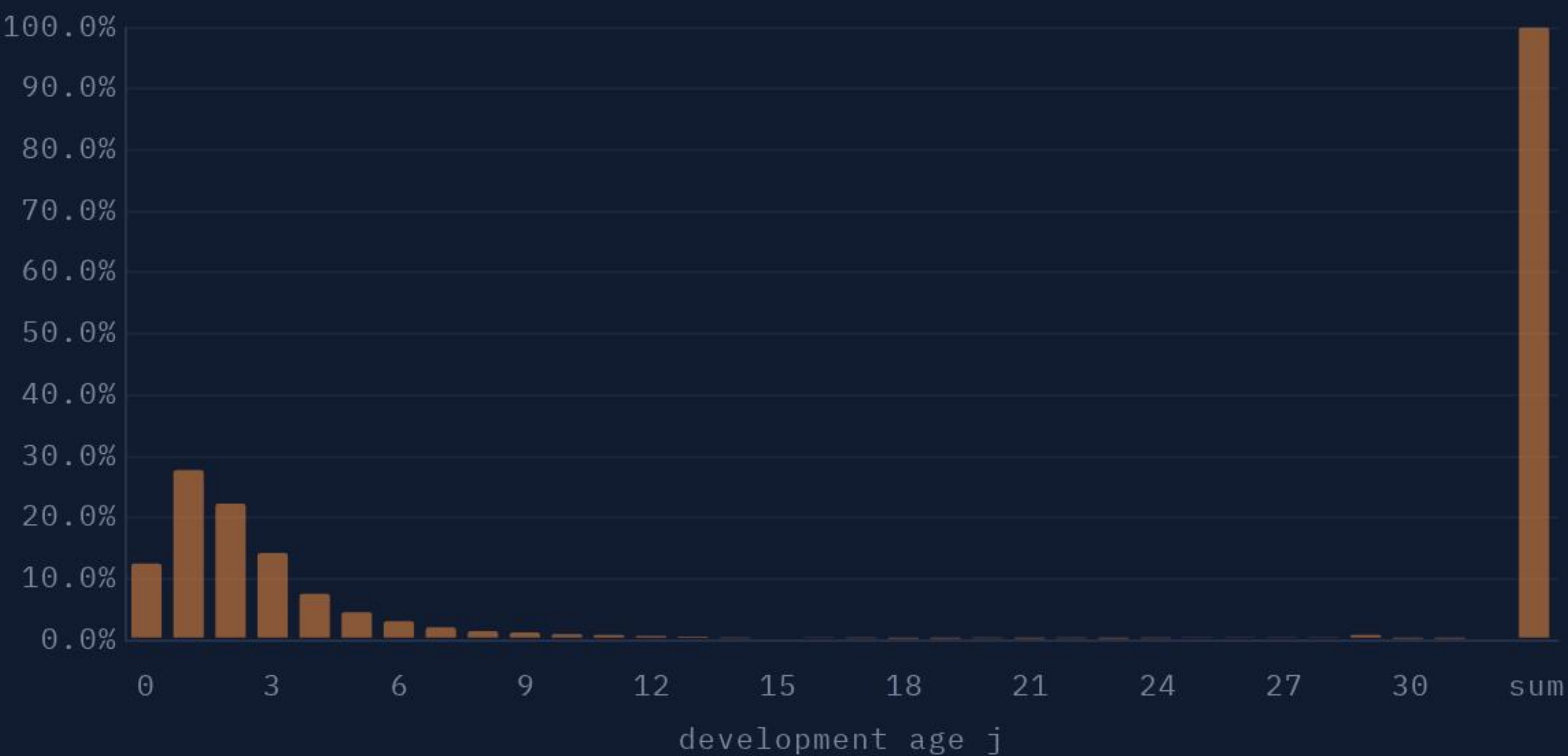


Fig. 3.4 MAD, undiscounted

IBNR by Accident Year





EXHIBIT 04

Summary & Method Comparison

All three methods start from the same paid-loss triangle (1993–2024, latest diagonal of MAD 2.76bn) and diverge only in how they extrapolate the unpaid tail. The comparison below quantifies how much that choice of method moves the reserve.

CHAIN LADDER – ULTIMATE MAD 3.24bn IBNR MAD 489.0m	LONDON CHAIN – ULTIMATE MAD 3.17bn IBNR MAD 409.5m	TAYLOR SEPARATION – ULTIMATE MAD 3.13bn IBNR MAD 369.7m	SPREAD ACROSS METHODS MAD 119.3m 3.8% of lowest estimate
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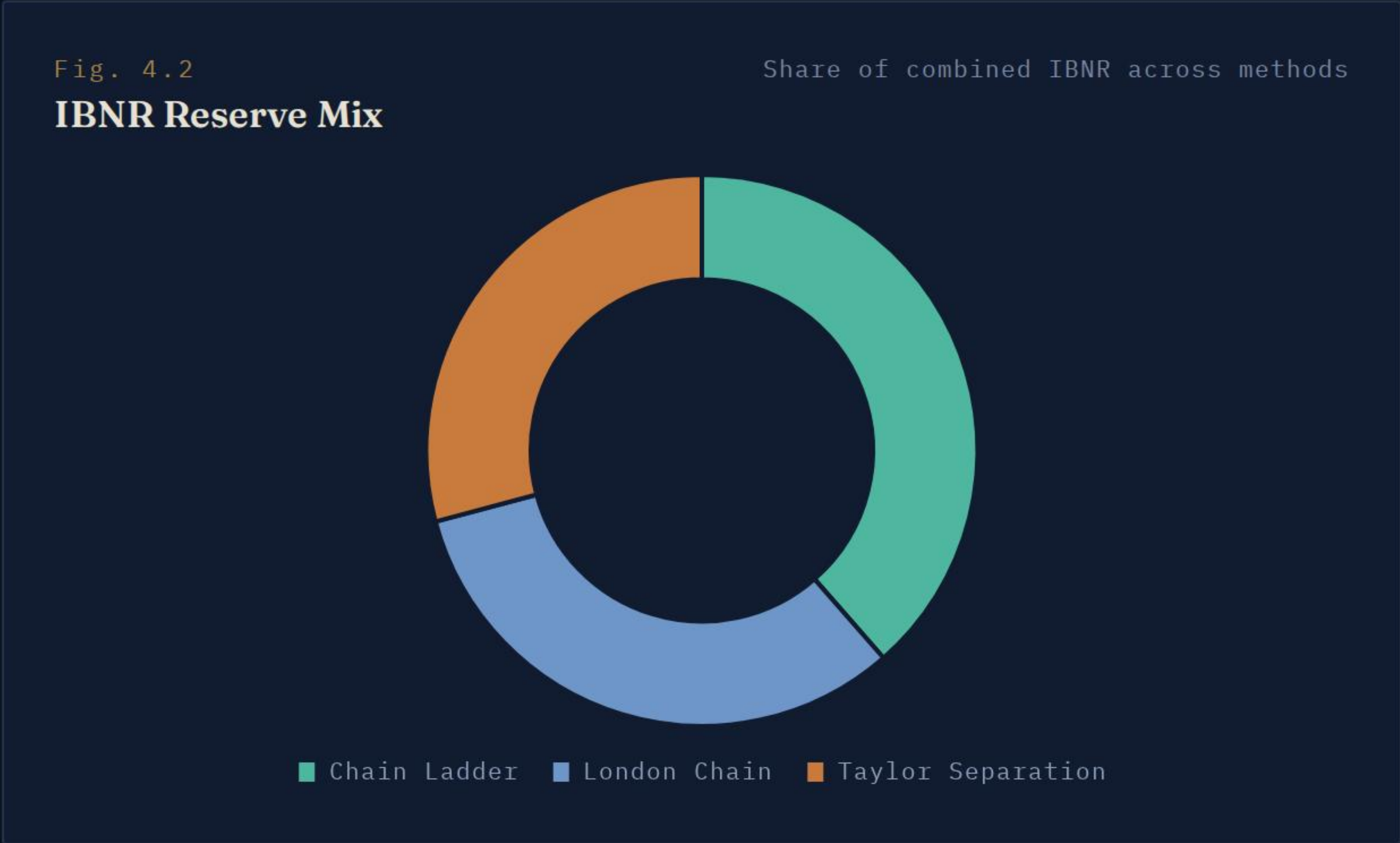
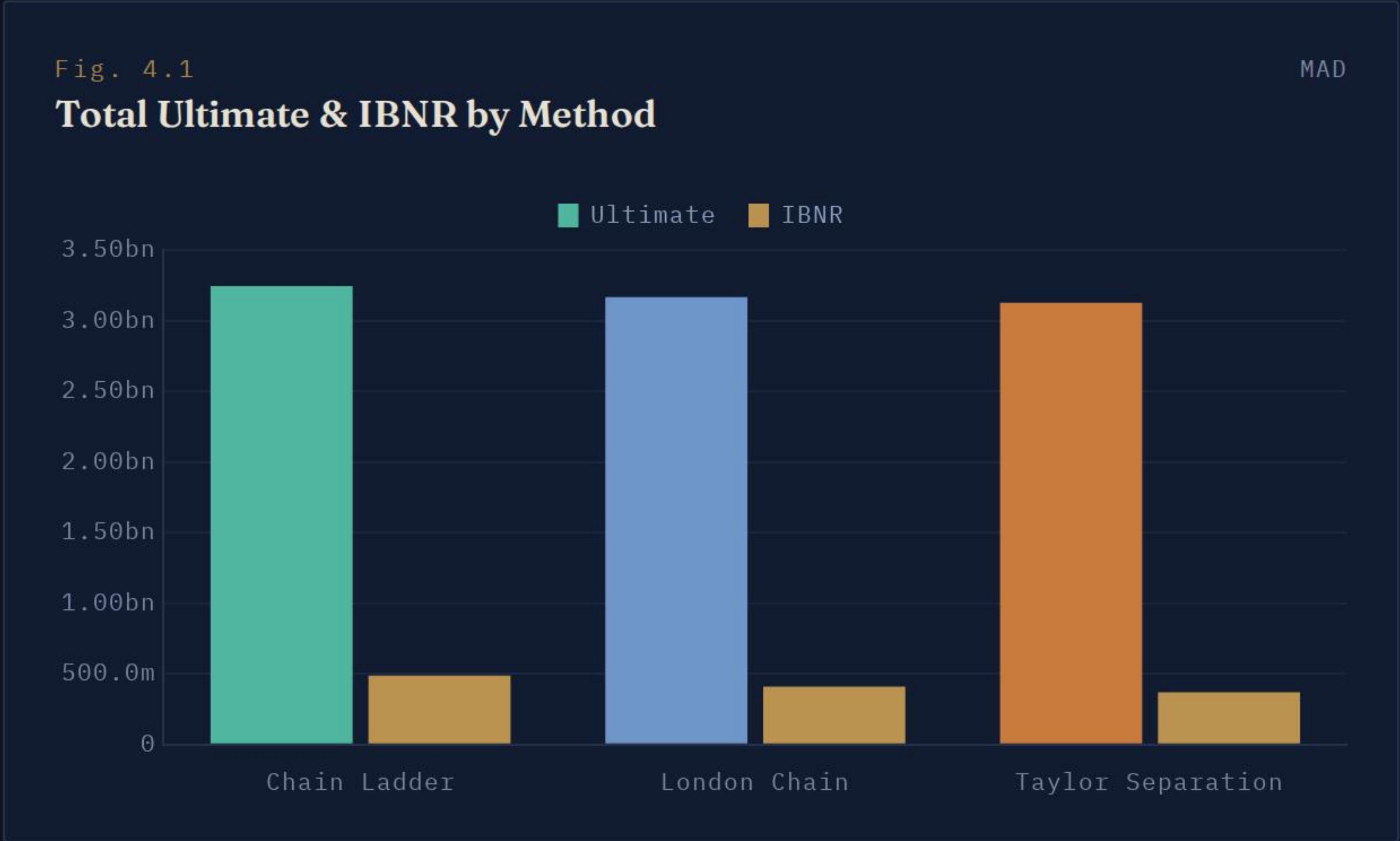


Fig. 4.3

Chain Ladder · London Chain · Taylor Separation

IBNR by Accident Year — Method Overlay

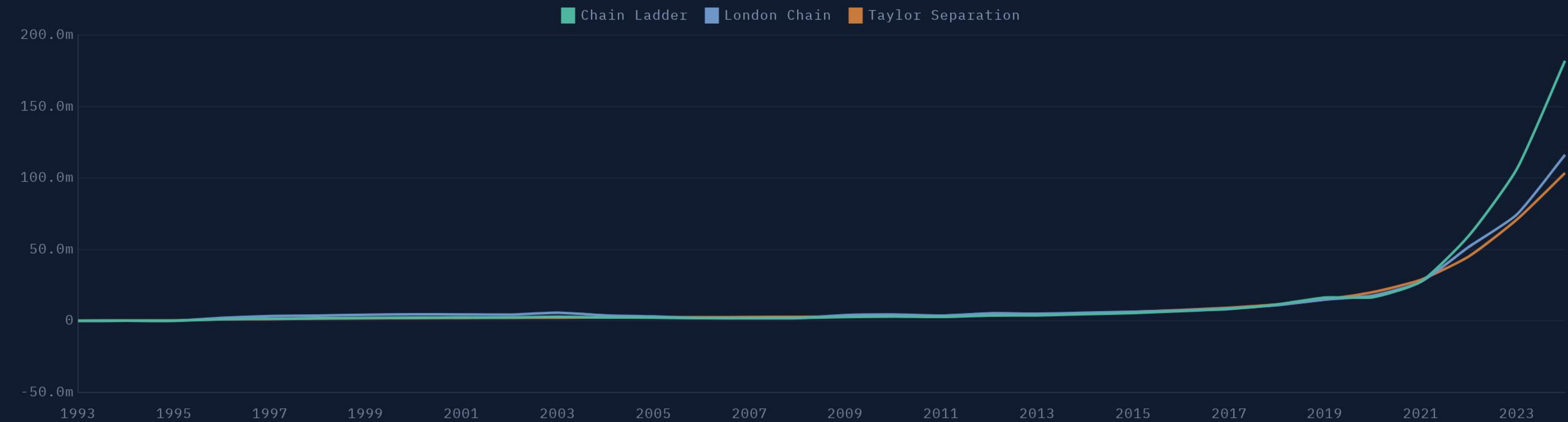
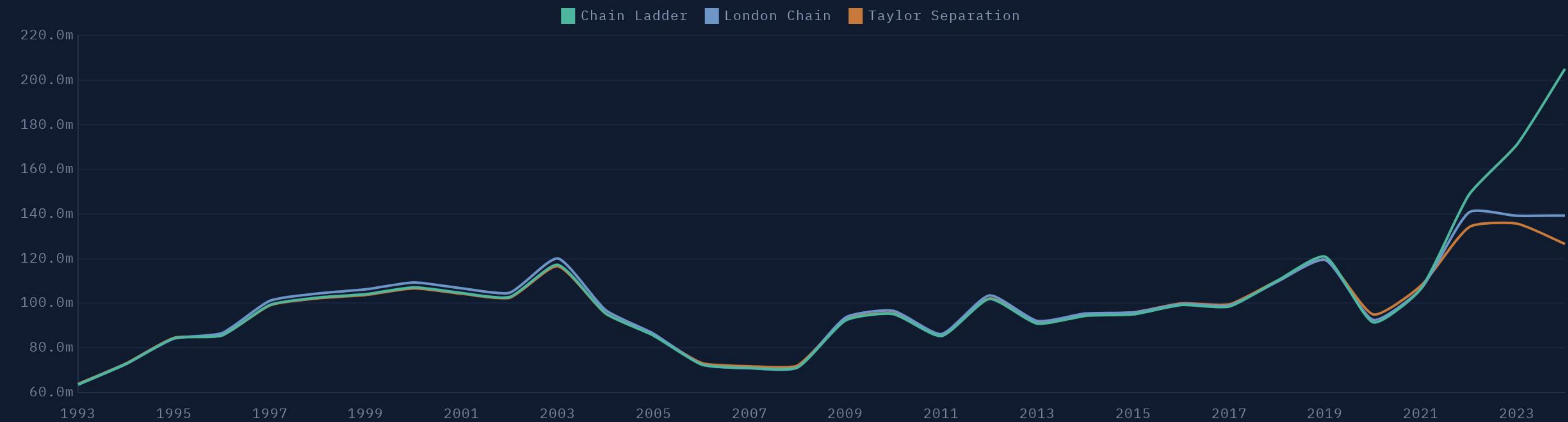


Fig. 4.4

Chain Ladder · London Chain · Taylor Separation

Ultimate Loss by Accident Year — Method Overlay



Ultimate & IBNR by Accident Year, All Methods

Accident Year	Paid to Date	Ultimate – CL	Ultimate – LC	Ultimate – Taylor	IBNR – CL	IBNR – LC	IBNR – Taylor
1993	MAD 63,489,194	MAD 63,489,194	MAD 63,489,194	MAD 63,742,420	MAD 0	MAD 0	MAD 253,226
1994	MAD 72,666,364	MAD 72,731,652	MAD 72,731,652	MAD 72,972,013	MAD 65,288	MAD 65,288	MAD 305,649
1995	MAD 84,090,881	MAD 84,229,248	MAD 84,088,624	MAD 84,445,318	MAD 138,367	MAD -2,257	MAD 354,437
1996	MAD 84,426,967	MAD 85,517,168	MAD 86,568,153	MAD 85,517,978	MAD 1,090,201	MAD 2,141,185	MAD 1,091,010
1997	MAD 97,729,057	MAD 99,211,924	MAD 101,079,679	MAD 98,994,733	MAD 1,482,867	MAD 3,350,622	MAD 1,265,676
1998	MAD 100,610,228	MAD 102,495,368	MAD 104,338,480	MAD 102,177,574	MAD 1,885,140	MAD 3,728,252	MAD 1,567,346
1999	MAD 101,916,438	MAD 104,053,479	MAD 106,194,223	MAD 103,694,022	MAD 2,137,041	MAD 4,277,785	MAD 1,777,584
2000	MAD 104,689,057	MAD 107,053,151	MAD 109,298,661	MAD 106,634,142	MAD 2,364,094	MAD 4,609,604	MAD 1,945,085
2001	MAD 102,159,371	MAD 104,569,394	MAD 106,636,856	MAD 104,223,939	MAD 2,410,023	MAD 4,477,485	MAD 2,064,568
2002	MAD 100,330,450	MAD 102,746,330	MAD 104,693,437	MAD 102,470,584	MAD 2,415,880	MAD 4,362,987	MAD 2,140,134
2003	MAD 114,348,860	MAD 117,203,325	MAD 120,017,484	MAD 116,603,330	MAD 2,854,465	MAD 5,668,624	MAD 2,254,470
2004	MAD 93,024,656	MAD 95,392,537	MAD 96,873,364	MAD 95,361,538	MAD 2,367,881	MAD 3,848,708	MAD 2,336,882
2005	MAD 83,186,246	MAD 85,422,852	MAD 86,292,876	MAD 85,702,285	MAD 2,236,606	MAD 3,106,630	MAD 2,516,039
2006	MAD 70,580,026	MAD 72,490,551	MAD 72,524,310	MAD 73,151,378	MAD 1,910,526	MAD 1,944,284	MAD 2,571,352
2007	MAD 69,046,402	MAD 70,966,938	MAD 70,837,349	MAD 71,730,531	MAD 1,920,535	MAD 1,790,947	MAD 2,684,129
2008	MAD 69,243,020	MAD 71,232,489	MAD 71,162,938	MAD 72,058,185	MAD 1,989,469	MAD 1,919,918	MAD 2,815,165
2009	MAD 89,516,484	MAD 92,226,109	MAD 93,509,672	MAD 92,526,495	MAD 2,709,625	MAD 3,993,188	MAD 3,010,010
2010	MAD 92,073,614	MAD 95,075,463	MAD 96,517,021	MAD 95,363,336	MAD 3,001,849	MAD 4,443,407	MAD 3,289,722
2011	MAD 82,475,429	MAD 85,263,157	MAD 86,142,427	MAD 85,935,462	MAD 2,787,728	MAD 3,666,998	MAD 3,460,034
2012	MAD 98,147,823	MAD 101,880,857	MAD 103,454,508	MAD 102,071,125	MAD 3,733,034	MAD 5,306,685	MAD 3,923,302
2013	MAD 86,976,454	MAD 90,810,410	MAD 91,973,967	MAD 91,530,407	MAD 3,833,957	MAD 4,997,514	MAD 4,553,953
2014	MAD 89,663,189	MAD 94,323,138	MAD 95,348,276	MAD 95,023,688	MAD 4,659,949	MAD 5,685,087	MAD 5,360,498
2015	MAD 89,493,517	MAD 94,973,104	MAD 95,852,890	MAD 95,787,131	MAD 5,479,586	MAD 6,359,372	MAD 6,293,614
2016	MAD 92,303,198	MAD 99,177,589	MAD 99,634,125	MAD 99,888,932	MAD 6,874,390	MAD 7,330,926	MAD 7,585,734
2017	MAD 90,248,684	MAD 98,486,869	MAD 98,971,175	MAD 99,436,428	MAD 8,238,184	MAD 8,722,490	MAD 9,187,744
2018	MAD 98,581,640	MAD 110,057,550	MAD 109,590,852	MAD 110,066,289	MAD 11,475,910	MAD 11,009,211	MAD 11,484,648
2019	MAD 104,453,213	MAD 120,837,243	MAD 119,393,213	MAD 119,420,032	MAD 16,384,030	MAD 14,940,000	MAD 14,966,819
2020	MAD 74,790,716	MAD 91,379,790	MAD 92,499,197	MAD 94,936,895	MAD 16,589,074	MAD 17,708,481	MAD 20,146,179
2021	MAD 79,065,032	MAD 106,641,657	MAD 106,497,009	MAD 107,897,248	MAD 27,576,626	MAD 27,431,977	MAD 28,832,217
2022	MAD 88,838,510	MAD 148,670,432	MAD 140,720,273	MAD 134,057,706	MAD 59,831,923	MAD 51,881,763	MAD 45,219,196
2023	MAD 64,614,533	MAD 171,151,537	MAD 139,187,043	MAD 135,631,418	MAD 106,537,004	MAD 74,572,510	MAD 71,016,885
2024	MAD 23,071,017	MAD 205,074,271	MAD 139,268,371	MAD 126,524,559	MAD 182,003,254	MAD 116,197,355	MAD 103,453,542
Total	MAD 2,755,850,270	MAD 3,244,834,777	MAD 3,165,387,299	MAD 3,125,577,120	MAD 488,984,507	MAD 409,537,029	MAD 369,726,850